

DATA ANALYSIS ACROSS BOUNDARIES

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MOTIVATION AND OBJECTIVES

This project addresses the gap in accessible tools for comparing data across Muslim-majority countries, particularly in Asia and Africa. The goal is to develop a user-friendly dashboard that visualizes key economic, social, and political indicators from trusted sources. By integrating multidimensional data, the dashboard will enable visual analysis, supporting informed decision-making and fostering collaboration.

METHODOLOGY

Data Collection

Sources


World Bank


OICStat


OECD

56 Muslim-majority countries (2000-2023)

Data Consolidation


Python scripts


SQL Database

Dashboard Building


Interactive Power BI dashboard

Economy

Health

Technology & Innovation

Agriculture

Climate

Business

Indicators

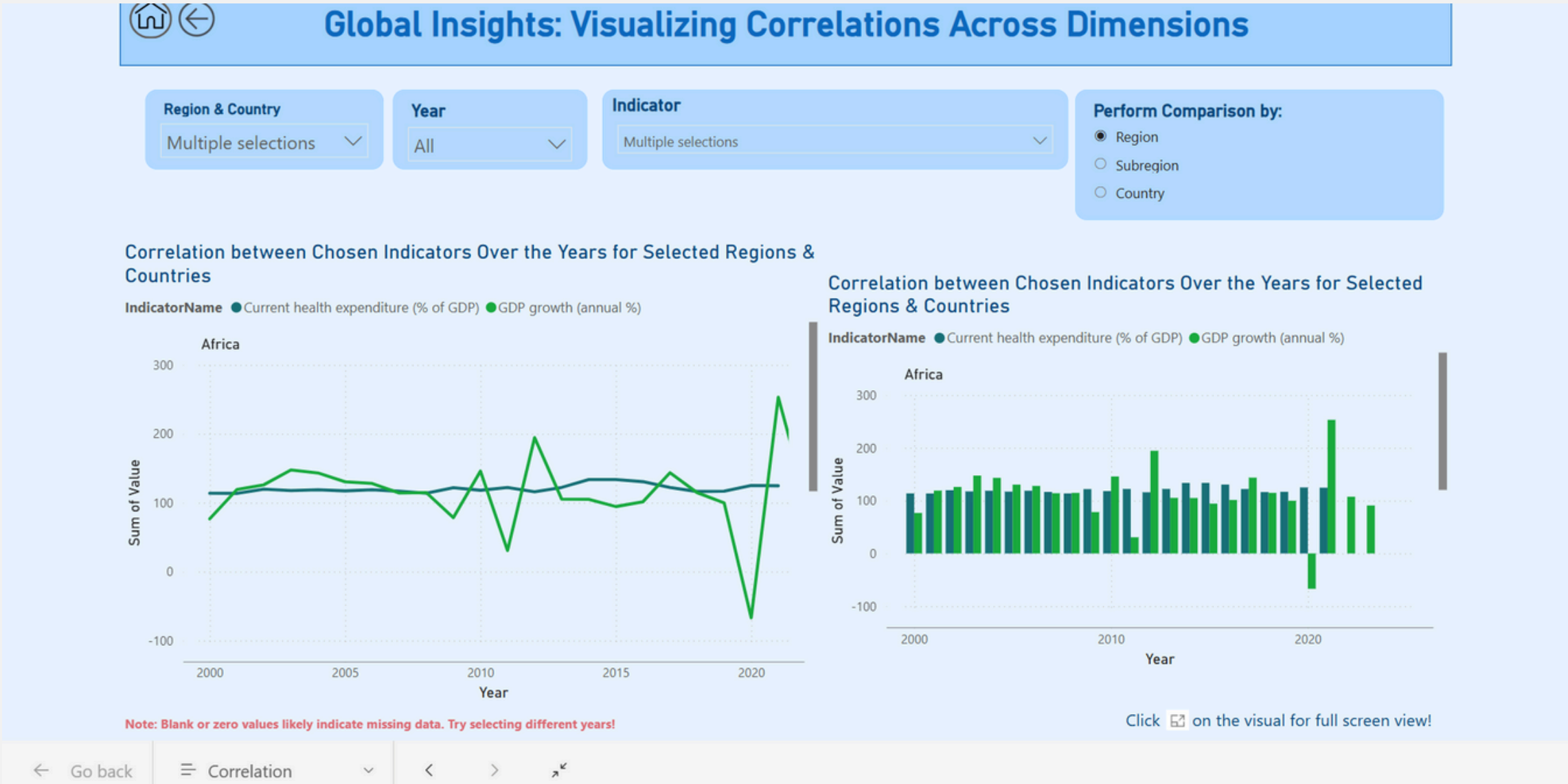
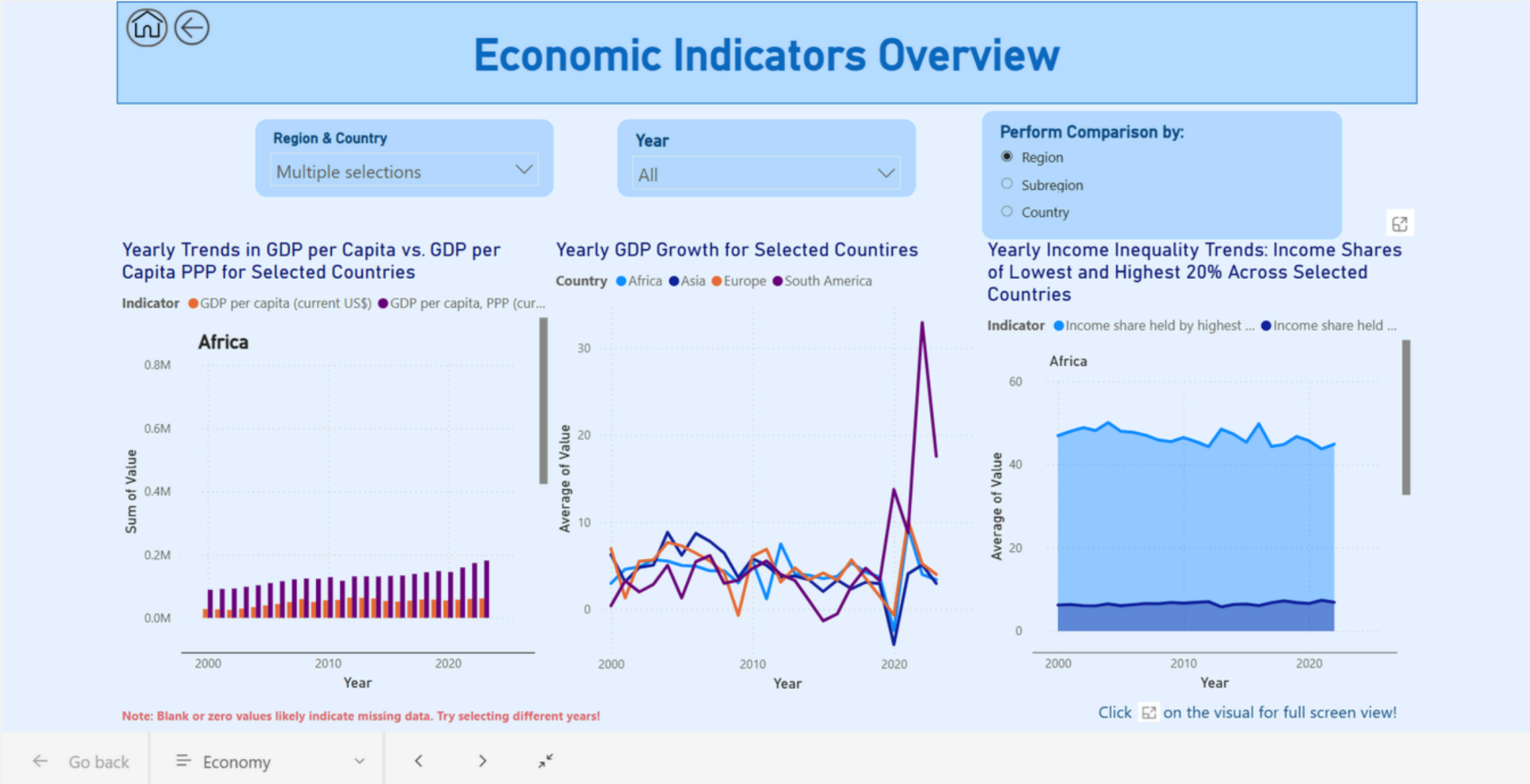
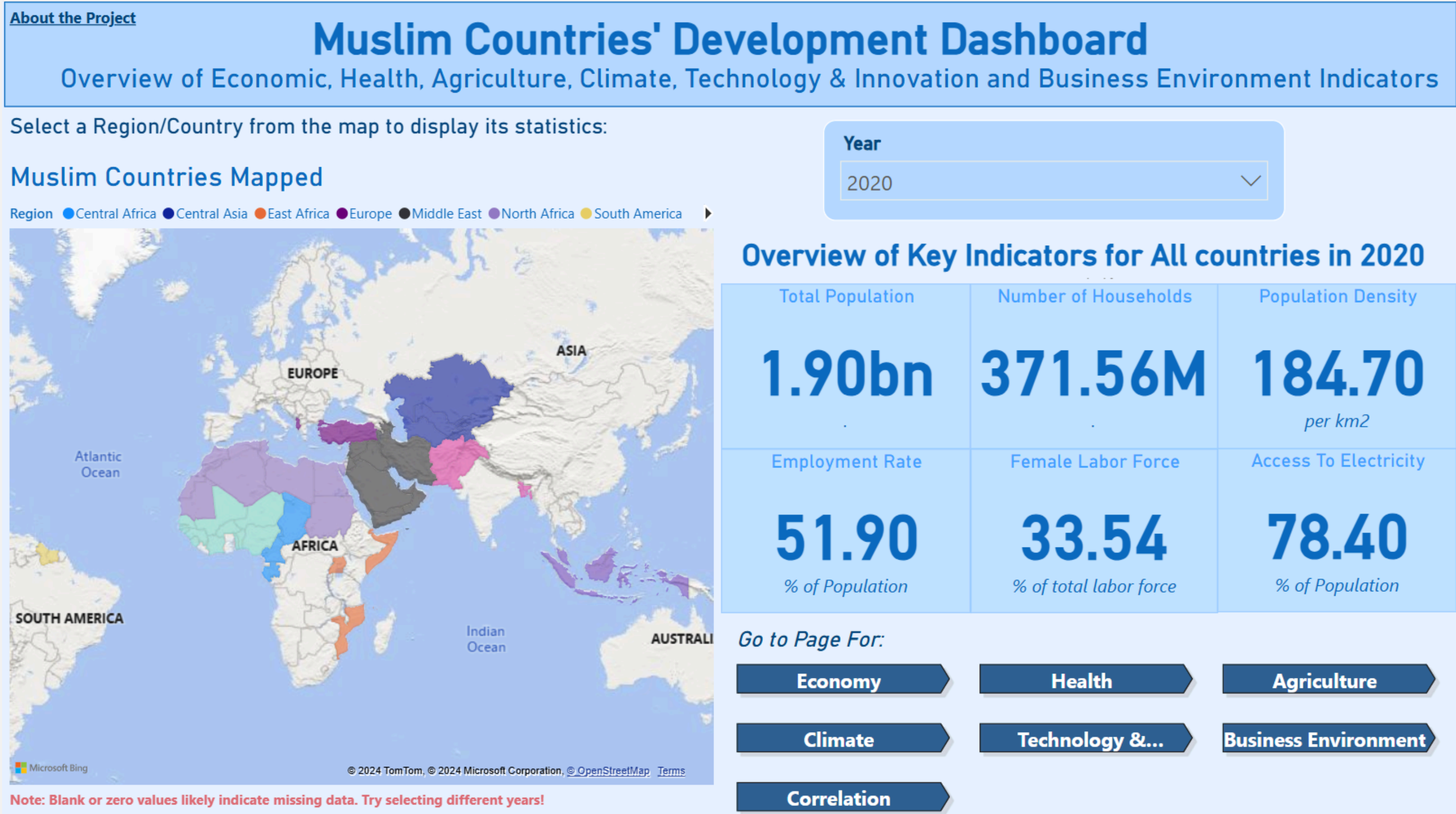
RESULTS

Database:

A clean, preprocessed database with 70+ performance indicators across eight dimensions for Muslim countries, compiled from reliable sources, is available at country and regional levels for detailed analysis.

Dashboard:

An eight-page interactive Power BI dashboard with slicers and filters allows for customized, user-friendly data exploration and visualization



LIMITATIONS AND FUTURE WORK

- Certain indicators are missing for small states (e.g., Palestine, non-OIC Muslim minorities).
- Recent years (2021–23) have patchy reporting in OICStat.
- Dashboard shows correlations, not causal inference.
- Uniform weighting of indicators may mask local priorities.

Future Directions:

- Include non-state Muslim populations (e.g. India, Ethiopia).
- Integrate GIS for spatial hotspot detection; add time-series forecasting (SARIMA exogenous).
- Conduct usability testing with policy teams; gather feedback to refine UX.
- Implement panel-data regressions or machine learning to tease out drivers of change.

CONCLUSION

This project consolidated data from 56 Muslim-majority countries and developed an interactive Power BI dashboard for multi-domain analysis. It offers a valuable tool for researchers and policymakers, enabling informed, data-driven decisions.

ACKNOWLEDGMENTS

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IMPLICATIONS

For Academics:

- A dedicated, centralized platform drastically lowers the barrier for regional comparative studies.
- Ready access to multi-dimensional data promotes cross-disciplinary research (health-economy-climate linkages).

For Policy-Makers:

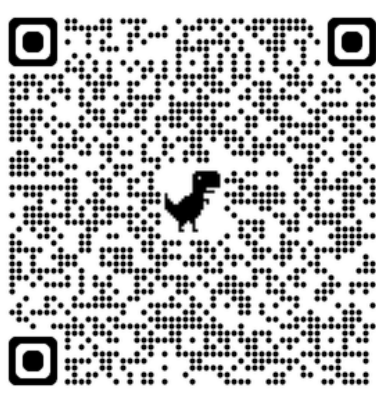
- Visualizing expenditure vs. outcome trade-offs can guide budget allocation.
- Interactive correlation module helps prioritize high-impact interventions.

For Practitioners & NGOs:

- Quickly identify climate-vulnerable or health-underinvested “hotspots.”
- Use the dashboard as a monitoring tool for SDG targets.

DASHBOARD

You may access our dashboard through the following QR code.



REFERENCES

Literature Review
Reference



Data Collection
Reference

